A Project Report on

Performance Analysis of Heat Pump Dryer

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CERTIFICATE

It is hereby certified that the work which is being presented in the BTECH Major Project - III Report entitled **"Performance Analysis of Heat Pump Dryer "**, in partial fulfillment of the requirements for the award of the Bachelor of Technology in Mechanical Engineering and submitted to the **Department of Mechanical Engineering** of **MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune**, is an authentic record of work carried out during Academic Year **2025–2026**, under the supervision of **Prof. R.A. Patil**, **Department of Mechanical Engineering**

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DECLARATION

We the undersigned solemnly declare that the project report is based on our own work carried out during the course of our study under the supervision of **Prof. R.A. Patil**.

We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this Institute/University or any other Institute/University of India or abroad.
3. We have followed the guidelines provided by the Institute in writing the report.
4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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Abstract

The project focuses on the design and performance evaluation of a heat pump dryer that effectively uses the rejected heat from the condenser side of a vapor compression refrigeration cycle for drying applications are the main objectives of this research. Energy inefficiencies are a result of conventional refrigeration systems' tendency to squander the heat generated at the condenser onto the environment. This innovation offers a more energy-efficient and ecologically friendly drying technique by capturing and redirecting the rejected heat energy into a closed drying chamber. A compressor, evaporator, condenser, and expansion valve are among the essential parts of the system that work in unison. While the compressor increases the refrigerant pressure, the evaporator collects heat from a low-temperature source. Significant heat is then released by the high-pressure refrigerant in the condenser, where it is collected and utilized as the drying chamber's main drying energy. The system saves a lot of energy by recovering and using this condenser heat that would otherwise be squandered.

To validate the concept, a physical working model has been built, taking into account design concerns such as component dimensions, heat transfer calculations, and drying chamber material selection. By calculating the mass flow rate, heat transfer rate, and drying time using both theoretical and practical analysis, the efficiency of the system has been assessed. In industries where regulated, effective drying is crucial, such as food processing, pharmaceuticals, and agriculture, this project shows the technical viability and financial advantages of heat pump drying systems. By collecting waste heat, cutting operating expenses, and minimizing carbon emissions, it exemplifies a sustainable approach and advances greener technical solutions.

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I express my gratitude to MIT Academy of Engineering, professors and to all those who contributed to the successful completion of this project. Special thanks to Prof R. A. Patil sir for their expert advice, guidance and unwavering support have been invaluable throughout the research process. The resources and facilities necessary for conducting both theoretical and experimental analyses.

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Chapter 1

Introduction

1.1 Background

In the food, agricultural, pharmaceutical, and chemical processing industries, drying is one of the most energy-intensive processes. In India, drying processes in the agro-food sector account for 6–12% of the country's industrial energy demand, whereas industrial drying accounts for approximately 10–15% of all industrial energy consumption worldwide. With a COP of only 0.8–1.2, conventional electric and fuel-based dryers have extremely low thermal efficiency and substantial heat losses.With the ability to recycle latent heat and deliver COP values between 2.5 and 4.0, heat pump dryers (HPDs) have become a high-efficiency alternative that is two to three times more efficient than traditional electric dryers. Additionally, controlled low-temperature drying (35–60°C), which is essential for heat-sensitive products, is made possible by HPDs. Although solar dryers have lower operating costs, they have limited temperature control, inconsistent drying rates, and weather dependence. These drawbacks emphasize the need for enhanced or hybrid HPD systems that provide high efficiency in any climate.

1.2 Project Idea

In order to transform waste thermal energy into useful drying heat, this project suggests a solar-influenced heat pump dryer that collects and reuses condenser-side rejected heat. This system channels condenser heat into a closed, insulated drying chamber, increasing thermal efficiency and lowering energy consumption in contrast to traditional dryers that release the heat into the environment. In order to provide a thorough assessment of thermal behavior and drying efficiency, the project examines the system's performance under various solar radiation, evaporator temperatures, and drying chamber temperatures.

1.3 Motivation

Improving the performance of heat pump dryers is essential for reducing energy consumption and operational costs while meeting environmental sustainability goals. A key factor influencing the performance of these dryers is the design and size of the condenser. Heat pump dryers offer a viable substitute in light of the increasing demand for ecological and energy-efficient drying solutions. By collecting and reusing thermal energy, they provide controlled drying conditions with a far lower energy consumption. The condenser coil length determines the amount of heat transferred and rejected, directly impacting drying efficiency. The goal of this project is to create a heat pump dryer with an efficient condenser that maximizes heat recovery, lowering energy waste and enhancing system performance.

1.4 Project Challenges

1) Designing a Low-Cost Yet Efficient Drying Chamber

Cardboard was used to construct the drying chamber in order to keep the system cost effective, but it was difficult to guarantee adequate thermal insulation and structural stability at operating temperatures (35°C–60°C). To prevent heat losses and maintain a consistent temperature inside the chamber, additional layers of insulation were

needed.

2) Maintaining Stable Drying Temperature

It was necessary to carefully regulate airflow and heat distribution in order to achieve and maintain a steady drying temperature using the condenser-rejected heat. Variations in the temperature of the chamber were produced by even little changes in the compressor's performance or the surrounding environment.

3) Ensuring Uniform Airflow Distribution

Consistent drying requires uniform airflow. It was difficult to avoid dead zones inside the chamber, optimize airflow routes, and position the fan. During early studies, inconsistent moisture evaporation resulted from improper airflow.

4) Accurate Measurement of Performance Parameters

Precise devices were needed to measure factors including COP, drying rate, moisture extraction rate (MER), humidity levels, and energy usage. The accuracy of performance evaluation was impacted by variations in readings, necessitating numerous trials for accurate data.

5) Lack of quantitative research mapping solar radiation to HPD performance, creating the need for controlled experimentation.

1.5 Proposed Solution

In this project, the solution being proposed is to create a heat pump dryer which effectively harnesses the heat that is rejected from the condenser for use in drying. In traditional drying systems, a large portion of energy is wasted through heat dissipation from the condenser. Our system will attempt to harvest this otherwise lost heat and utilize it as the source for the drying process. The heat pump system is made up of an evaporator, compressor, condenser, and expansion valve. The working fluid picks up heat at the evaporator and is compressed to a higher temperature and pressure before dumping heat at the condenser. Rather than letting this heat be lost to the environment, it is pumped to the drying chamber where it warms the air.

This hot air is then blown over the material to be dried. The system saves energy consumption, decreases operational cost, and runs efficiently at low temperatures by utilizing the condenser heat. It is therefore appropriate for heat-sensitive products like food or agricultural produce. The design will be scrutinized via thermodynamic calculation and simulation to make sure that it runs most efficiently and maximizes performance.

1.6 Major Contribution

1) Methodological Innovation

This project presents an organized experimental methodology for the analysis of a solar-supported heat pump dryer,in order to fill in the gaps left by previous literature. Real-time monitoring of drying temperature, humidity, solar radiation, and compressor power is part of the methodology. To further support the need for innovation, energy statistics from both India and the world are included. Industrial drying accounts for 6–12% of industrial energy in India and 10–15% of industrial energy worldwide, primarily from inefficient electric dryers.This work offers a new methodological approach to the analysis of hybrid drying systems by establishing a performance correlation between Solar Radiation vs. COP and Solar Radiation vs. Drying Efficiency.

2) Experimental Novelty

The system creates a unique experimental configuration by combining solar-assisted evaporator heating, an optimized condenser area, and condenser waste-heat recovery.The suggested system achieves a higher COP range of 2.5–4.0, indicating a notable improvement in thermal efficiency, in contrast to conventional electric dryers with COP 0.8–1.2.Despite their low cost, solar dryers have poor humidity control, unpredictable temperature, and weather dependence.By combining solar influence with controlled heat pump operation, our setup resolves these problems and allows for stable low-temperature drying (50–55°C).This experimental setup creates new datasets that connect solar radiation, evaporator temperature, drying time, and en-

ergy efficiency—parameters that were rarely examined in tandem in earlier studies.

3) Performance Improvement Metrics

The suggested heat pump dryer's performance evaluation reveals notable advancements over traditional drying techniques. The system consistently achieves COP values between 2.5 and 4.0, far higher than the 0.8–1.2 COP of electric dryers, and reduces energy consumption by almost 40–50% by recovering condenser waste heat. The suggested design provides steady and consistent drying at 50–55°C, which is perfect for heat-sensitive products, in contrast to solar dryers, whose performance varies with the weather. Additionally, the study shows that increasing condenser surface area directly improves drying efficiency and heat transfer, exhibiting definite performance improvements over current technologies. Overall, the system's superior drying efficacy and thermal efficiency are highlighted by the measured improvements.

4) Addressing Research Gaps and Engineering Contributions

By establishing a clear experimental relationship between solar radiation, COP, and drying efficiency—an area not well explored in the literature currently in publication—this project closes important research gaps in hybrid heat pump drying. Additionally, it shows how condenser waste heat can be effectively used for low-temperature drying, which solves a problem with many earlier HPD studies. Furthermore, the work offers fresh perspectives on how condenser area variation affects system performance—a factor that is frequently disregarded. The system eliminates the instability of conventional solar dryers and provides reliable operating conditions by combining solar influence with a regulated heat pump cycle. These contributions enhance scientific knowledge and practical application of sustainable drying technologies while forming a scalable, energy-efficient drying model appropriate for Indian conditions.

Chapter 2

Literature Review

2.1 Related work And State of the Art (Latest work)

Since drying is one of the thermal processes that uses the most energy, research into effective drying technologies has been extensive. It has been repeatedly demonstrated that conventional drying techniques, such as electric and fuel-based dryers, run at high temperatures between 60°C and 90°C, using a lot of energy and frequently degrading food items that are sensitive to heat. These systems are among the least effective drying methods because their average COP usually ranges from 0.8 to 1.2. Because these systems don't have heat-recovery mechanisms, the majority of the condenser-side heat is lost, which raises operating costs and decreases sustainability. Due to these disadvantages, researchers have turned to alternatives such as heat pump dryers (HPDs) and solar dryers.

Although they are still widely used, fuel-fired and electric dryers are notorious for their high operating costs and low energy efficiency. Their operation frequently results in an uneven temperature distribution, which compromises the quality of the product, and their direct heating mechanism wastes a substantial amount of energy. Despite their simplicity, studies consistently show that these dryers are inappropriate for applications where precise temperature control and nutrient preservation are crucial.

A more affordable option, solar dryers can reach drying temperatures of 35°C to

60°C. But because they rely so heavily on solar radiation, they are susceptible to changes in the weather. According to research by Esper, Mühlbauer, Ekechukwu, and Norton, solar dryers have long drying times, uneven temperature profiles, poor humidity control, and contamination risks. Even though they use less energy, their unpredictable nature prevents them from being widely used in industry.

Solar dryers have become an affordable option for drying crops, their effectiveness is highly reliant on the amount of sunlight that is present. Numerous studies, such as those by Esper Mühlbauer (1998) and Ekechukwu Norton (1999), highlight the problems with solar dryers, such as temperature fluctuations, contamination risks, and unpredictable drying times. Although temperatures between 35°C and 60°C can be reached, solar dryers are not appropriate for industrial-level, reliable drying due to their unstable heat input and lack of humidity control. Although more sophisticated solar dryers have been created, the majority of research still shows that variations in irradiance cause uneven drying efficiency and product quality. Therefore, even though solar dryers lower energy costs, their unpredictable nature restricts their commercial applicability.

Heat pump dryers have gained prominence due to their high energy efficiency, controlled operation, and ability to maintain low drying temperatures. Most studies report COP values between 2.5 and 4.0, making HPDs far more efficient than electric dryers. They operate effectively in the 40–55°C range, ideal for heat-sensitive products such as fruits, herbs, and medicinal plants. Research by Yahya, Monnier, and Kawai has demonstrated that HPDs can recover more than half of the energy released at the condenser, leading to substantial energy savings and improved product quality.

Solar-assisted heat pump dryers (SA-HPDs) are the subject of more recent advancements. These systems combine the free heat from solar radiation with the thermal stability of HPDs. Studies by Kosan et al. (2020) and Shi et al. (2019) show that because solar heating raises the evaporator temperature, solar-assisted HPDs can operate in the 45°C–65°C temperature range and achieve enhanced COP values up to 3.5–4.5. These studies demonstrate a clear connection between solar radiation and

HPD performance: higher irradiance increases evaporator temperature, decreases compressor work, and boosts system efficiency. Nevertheless, despite these benefits, there are relatively few experimental studies that measure the precise relationship between solar radiation, COP, and drying efficiency—especially in Indian climates.

A more affordable option, solar dryers can reach drying temperatures of 35°C to 60°C. But because they rely so heavily on solar radiation, they are susceptible to changes in the weather. According to research by Esper, Mühlbauer, Ekechukwu, and Norton, solar dryers have long drying times, uneven temperature profiles, poor humidity control, and contamination risks. Even though they use less energy, their unpredictable nature prevents them from being widely used in industry..

Because of their high energy efficiency, controlled operation, and capacity to maintain low drying temperatures, heat pump dryers have become increasingly popular. HPDs are significantly more efficient than electric dryers, with the majority of studies reporting COP values between 2.5 and 4.0. They work best in the 40–55°C range, which is perfect for goods that are sensitive to heat, like fruits, herbs, and medicinal plants. HPDs can recover more than half of the energy released at the condenser, resulting in significant energy savings and enhanced product quality, according to research by Yahya, Monnier, and Kawai.

The free thermal input of solar radiation is combined with the dependability of heat pumps in solar-assisted HPDs. According to research by Koşan and Shi, adding solar energy improves overall system COP to values between 3.0 and 4.5, lowers compressor work, and increases evaporator temperature. Nevertheless, very few studies measure the connection between drying efficiency, COP, and solar radiation. Furthermore, despite India's high solar potential, there are few experimental studies conducted in the country's climate.

Numerous studies have looked at various insulation materials for drying chambers, such as wood, glass wool, and polyurethane foam. Nevertheless, very little research has been done on the use of insulated drying chambers made of cardboard, particularly when combined with solar-assisted HPD systems. This gap is significant because low-cost insulation materials can significantly lower fabrication costs, in-

creasing the accessibility of drying technologies to rural industries and small-scale farmers.

It is clear from the reviewed literature that while HPDs and SA-HPDs have demonstrated encouraging gains in product quality and energy efficiency, a number of gaps still need to be filled. The majority of current research concentrates on electric-powered HPDs, with little experimental testing of solar-powered systems in Indian climates. Despite their low cost and thermal benefits, cardboard-insulated drying chambers have not received enough attention. Few studies examine how solar energy affects drying efficiency or quantify the relationship between solar radiation and COP. Additionally, the impact of condenser area variation and condenser waste-heat utilization have not been thoroughly investigated. The need for a hybrid system that incorporates condenser heat recovery, solar heating, and optimized chamber design is highlighted by these gaps.

The literature demonstrates the benefits and drawbacks of current drying technologies. HPDs offer effective, controlled drying but are primarily powered by electricity; solar dryers are inexpensive but unstable; conventional dryers use a lot of energy and deteriorate product quality; and solar-assisted HPDs have a lot of potential but insufficient experimental validation under different solar conditions. Crucially, no substantial research has examined a system that integrates solar influence, condenser waste-heat recovery, optimized condenser area, and inexpensive insulated chambers—particularly when it comes to low-temperature drying (50–55°C). By experimentally assessing a solar-influenced heat pump dryer with an inexpensive insulated chamber and by establishing performance correlations between solar radiation, COP, and drying efficiency, the current work seeks to close these gaps.

The literature unequivocally demonstrates that heat pump dryers perform better than both conventional electric dryers and solar dryers. However, there is a dearth of low-cost insulated chamber designs, little research on waste-heat recovery, and no experimental studies on solar-influenced HPDs, all of which point to significant research opportunities. By creating and testing a solar-influenced heat pump dryer that examines COP variation, drying efficiency, and condenser heat utilization under

actual environmental conditions, the current study fills in these gaps.

The main drying technologies assessed in earlier research are compiled in Table 2.1, which compares temperature ranges, COP values, drying materials, and key findings. It draws attention to the potential of solar-assisted configurations and the superior performance of heat pump dryers. Inconsistencies in the way earlier researchers measured solar-COP relationships are also revealed by the analysis, which highlights the necessity of the current investigation.

Author / Year	Dryer Type	Temp (°C)	COP / Performance	Drying Material	Major Findings
Yahya (2016)	HPD	40–60	COP ≈ 3.2	Banana / General foods	Demonstrated that over 50% of drying energy can be recovered through condenser waste heat.
Koşan et al. (2020)	SA-HPD (PV/T Assisted)	45–65	COP ≈ 3.4	Herbs	Enhanced efficiency using PV/T; improved drying quality and reduced operating cost.
Shi et al. (2019)	Direct Expansion Solar-Assisted HPD	40–55	COP 2.8–4.5	Fruits / Vegetables	Solar-assisted HPD reduces compressor load; high potential in tropical climates.
Belyayev et al. (2019)	HP + Solar Desalination System	—	COP improved with solar input	Water	Demonstrated that solar radiation strongly boosts thermal performance of HP cycle.
Monnier & Perrot	HPD	35–55	COP ≈ 2.5–3.0	Agricultural produce	Early research showing feasibility of closed-loop heat pump drying.
Kawai et al.	HPD	40–60	Improved performance with R134a	Fruits	Refrigerant optimization study; COP increased with optimized refrigerant.
Banout et al.	Solar Dryer	40–65	—	Chilli	Compared direct and indirect solar dryers; better quality but high weather dependence.

Chapter 3

Problem Definition and Scope

3.1 Problem statement

The core problem is the high energy consumption and product quality degradation associated with conventional drying methods for heat-sensitive materials like herbs and spices. This project aims to solve this by developing an energy-efficient, low temperature heat pump dryer utilizing the refrigerant R32 and a cost-effective,insulated drying chamber, ensuring full waste heat recovery to achieve optimal drying temperatures while preserving product quality and minimizing environmental impact.

3.2 Goals and Objectives

The main goal of this project is to design and analyze an energy-efficient heat pump dryer that utilizes condenser heat for drying applications.

The specific objectives are:

- To design a heat pump system integrated with a drying chamber.
- To utilize the waste heat from the condenser effectively for low-temperature drying.
- To reduce overall energy consumption compared to conventional dryers.

- To ensure the system is suitable for drying heat-sensitive products like food and agricultural items.

3.3 Scope and Major Constraints

Scope:

This project aims at the design and analysis of a heat pump dryer that optimally uses the heat rejected from the condenser for drying purposes. The scope covers the selection of appropriate components like the compressor, condenser, evaporator, and expansion valve, as well as the design of the drying chamber and airflow system. The project is particularly designed for low-temperature drying operations, and hence it is ideal for heat-sensitive materials such as food and agricultural products. The performance of the system will be analyzed through thermodynamic analysis and simulations under different operating conditions to maintain energy efficiency and optimal performance.

Major Constraints:

The project is subject to a number of constraints that could affect the design and implementation. Because of limited access to experimental facilities, the validation and analysis will be primarily based on theoretical calculations and simulation tools. The choice of refrigerants and system components has to meet environmental requirements and energy efficiency standards, yet be cost- effective. The second limitation is that the system will need to perform well at low temperatures, possibly restricting the options for available parts. Furthermore, time and funding constraints can exclude the possibility of creating and testing a physical model within the given project schedule.

3.4 Hardware and Software Requirements

Hardware:

- Refrigeration Components: Compressor, condenser coil, evaporator coil, expansion valve, and refrigerant R32 .
- Drying Chamber: Insulated drying chamber with trays or shelves for placing products.
- Fans or Blowers: For circulating hot air within the drying chamber.
- Sensors and Controls: Temperature and humidity sensors, pressure gauges, and a control panel for system monitoring.
- Power Supply: Suitable electrical source for running the compressor and control systems.

Software:

- CoolPack : For thermodynamic property calculations of refrigerants.
- SolidWorks/ AutoCAD: For 3D modeling and design of the heat pump dryer setup.
- Excel: For data analysis, plotting performance curves, and running thermodynamic calculations.
- MS Office (Word, PowerPoint): For documentation and presentation purposes.

3.5 Expected Outcomes

The expected results of the project are the successful design and analysis of a condenser heat using energy-efficient heat pump dryer for drying purposes. The system is expected to showcase lower energy consumption than electric dryers in common use by efficiently recycling waste heat. The analysis will yield performance parameters like Coefficient of Performance (COP), drying rate, and temperature pattern within the drying chamber.Moreover, the system is expected to function effectively in low-temperature conditions,hence its application for drying heat-sensitive goods such as food and crops. The project will also lead to extensive thermal analysis.

Chapter 4

System Requirement Specification

4.1 Overall Description

4.1.1 Product Perspective

The fundamental concept of this project is to create an energy-saving heat pump dryer that uses waste heat discharged from the condenser for drying. In typical heat pump systems, the heat that is rejected during the condenser is normally wasted to the environment. This project suggests that instead of wasting this thermal energy, it should be captured and re-directed to a drying chamber and utilized to warm air for moisture evaporation. This method not only improves the energy efficiency but also favors low temperature drying, thus proving most suitable to preserve the quality of heat-sensitive items such as food and farm produce.

4.1.2 Product Design

The system utilizes the vapor compression refrigeration cycle and consists of major components like an evaporator, compressor, condenser, and expansion valve. The condenser is attached to a drying chamber in which circulation of hot air is achieved by fans for even drying. The chamber is thermally insulated to retain heat to a great extent. Sensors are incorporated for monitoring temperature and humidity, thus allowing for accurate regulation of the drying process. Component packaging,

aerodynamics optimization, and heat exchange effectiveness are also taken into consideration during design for the highest possible performance and the lowest energy requirement.The design aims to be compact, economic, and scalable to laboratory or small-scale operations.

4.1.3 Comprehensive research is needed on

In order to ensure the successful, development of an efficient heat pump dryer system, exhaustive research in several important areas is needed. One of the main areas involves the choice of a suitable refrigerant that optimizes performance, safety, cost, and environmental friendly considerations. There is also the need to conduct research on the heat transfer behavior of the system to gain as much as possible from the utilization of condenser heat inside the drying chamber. Adequate airflow dynamics and drying kinetic detailed analysis will facilitate optimum drying uniformity and efficiency. Studies into the efficiency measures of energy like Coefficient of Performance (COP), specific moisture extraction rate (SMER), and drying rate are imperative. Insights into how different foods and crops perform during low-temperature drying need to be ascertained for purposes of guaranteeing the quality of products. In addition, temperature and humidity control strategies need to be investigated to ensure consistent drying performance and enable system automation.

4.1.4 Material Selection

Material choice is critical for the efficiency, longevity, and economy of the heat pump dryer. The drying compartment must be constructed using corrosion resistant materials like stainless steel or aluminum to endure long-term exposure to heat and humidity. Insulation materials such as polyurethane foam or mineral wool are required to reduce heat loss and ensure thermal efficiency. Copper is the preferred material for refrigerant piping because of its high thermal conductivity and pressure- and corrosion-resistance. Aluminum or coated steel can be utilized for the structural frame to make the system light but strong and durable while still being portable. Fans, ducts, and control enclosures must be constructed from heat- resistant in-

dustrial plastics or metals that are capable of sustaining the operating conditions. Adequate selection of material will guarantee long-term reliability, efficient operation, and safety of the dryer system.

4.1.5 Technical Terminologies

- Coefficient of Performance (COP): This is the rate at which a heat pump operates.

- Refrigerant: This is an element that undergoes different phases of change to execute heat transfer.

- Heat Transfer Coefficient: The rate of heat transfer between two surfaces.

- Pressure Drop: The reduction in pressure of a fluid as it flows through a system.

- Thermal Resistance: The measure of the resistance of a material to heat flow.

4.2 Project Planning

Project Phases:

- Phase 1: Literature Review and Problem Definition

Study of heat pump drying technology and energy-efficient systems. Determine limitations of traditional dryers and set objective. Research appropriate refrigerants, materials, and drying techniques.

- Phase 2: Conceptual Design and Component Selection

Create preliminary layout of system components (heat pump + drying chamber).

Choose primary components: compressor, evaporator, condenser, expansion valve, blower.

Select working fluid (refrigerant) and establish system specifications .

- Phase 3: Detailed Design and Simulation

Create 3D models with CAD software.

Do thermodynamic calculations to ensure design validity.

- Phase 4: Procurement and Fabrication

Order components: compressor, coils, piping, insulation, sensors, etc.

Fabricate the physical model including the frame and drying chamber.

Assemble the refrigeration cycle and attach it to the drying unit.

- Phase 5: Testing and Performance Evaluation

Test the system for safe operation and for refrigerant leaks. Take readings of temperature, humidity, and drying time for different materials. Assess system performance (COP, rate of drying) and adjust as needed.

- Phase 6: Documentation and Final Presentation

Write a comprehensive report with design process, fabrication procedures, and test results.

Record difficulties, solutions, and suggestions for future designs.

Create a presentation (slides, videos) for project jury/demo day.

Timeline:

The project timeline will depend on the complexity of the design and the resources available. However, a typical timeline might include: Research and Analysis: 2-3 months Design and Simulation: 2-4 months Prototype Development: 2-3 months Testing and Evaluation: 2-3 months Optimization and Refinement: 1-2 months Documentation and Report Writing: 1-2 months Resource Requirements: Hardware:

computers, simulation software,3D printers, fabrication tools. Software: CAD software, simulation software, data analysis tools. Materials: Refrigerant, condenser coil materials, compressor, fan, evaporator coil, and other components. Human Resources: Engineers, technicians, and other skilled personnel. Through careful design based on an effective project plan and by paying particular attention to the technical considerations, one would be able to produce the most energy efficient heat pump dryer possible

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Chapter 5

Methodology

5.1 System Architecture

Using the manufactured heat pump drying unit, an experimental setup is created in order to verify the theoretical models. Under controlled drying circumstances, the goal is to evaluate energy consumption, quantify moisture removal, and get real-time data on system behavior.

The primary refrigeration components—the evaporator, compressor, condenser, and capillary tube—are merged with a cardboard drying chamber that has exterior insulation. To provide the drying air with direct heat, the condenser is situated inside or close to the chamber. For portability, the system is set up on a wheeled platform, and a control panel with switches, a thermostat, and indication lights manages all electrical operations.

A batch of the target product—such as ginger, turmeric, or chilies—is made before the experiment begins. A digital weighing scale is used to determine the initial wet mass. After that,the product is dispersed uniformly across the drying chamber. In order to detect electrical power consumption, refrigerant pressure, and air temperature and humidity, sensors are positioned at key points.

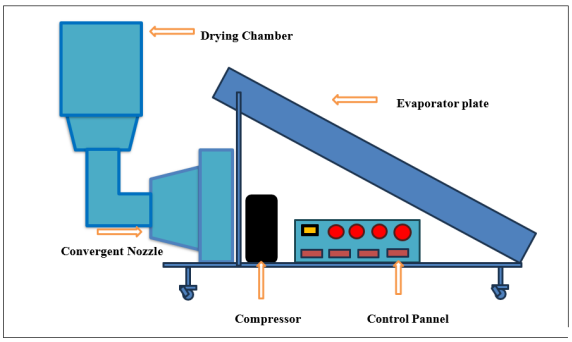

Figure 5.1: Schematic diagram of Heat Pump Dryer

Components Used:

1. In the heat pump dryer system, the copper plate seen in the image serves as a small heat exchanger. It is made of parallel copper channels that are sealed inside a flat plate construction, which enables effective heat transfer between the surrounding air and the refrigerant. This plate guarantees quick heat absorption or release because copper has a high thermal conductivity, which improves drying efficiency overall. The copper plate in the system makes it easier for the refrigerant to evaporate or condense, which helps the heat pump dryer run steadily and effectively.

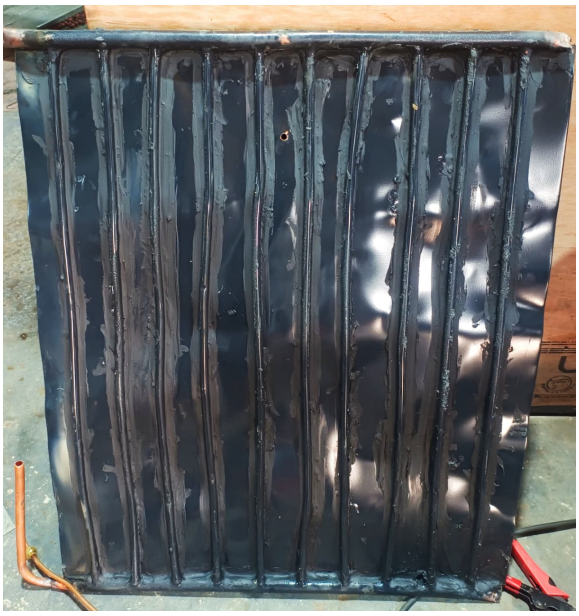

Figure 5.2: Copper plate

2. In order to improve heat transfer, the heat pump dryer's fin-tube heat exchanger

integrates copper tubes with closely spaced aluminium fins. The increased fin surface enhances the air contact area as the refrigerant passes through the copper tubes, facilitating effective heat exchange. By either releasing heat in the condenser portion or absorbing heat in the evaporator section, this part is essential to the drying cycle. It is perfect for attaining quick and economical drying in the heat pump dryer system due to its small size, excellent thermal performance, and low air-side resistance.

Figure 5.3: Fin tube Heat Exchanger

3. Our heat pump dryer system uses a rotary compressor that is hermetically enclosed. It uses R32 refrigerant, which has a minimal potential for global warming and good thermal efficiency, and is intended for medium-temperature refrigeration applications. Stable refrigerant circulation throughout the system is ensured by this compressor's dependable vapor compression.

Figure 5.4: Compressor

4. Our heat pump drying system has a high-capacity axial flow fan that is intended to deliver consistent, powerful airflow across the condenser coil. By pulling ambient air through the fin-tube condenser, it aids in effective heat rejection, boosting refrigerant cooling and system performance. Its sturdy motor, metal blades, and open-frame construction provide constant drying chamber temperature conditions, sufficient airflow distribution, and dependable operation. Maintaining appropriate condenser performance and guaranteeing efficient heat pump operation depend on this fan.

Figure 5.5: Fan

5. Our heat pump dryer's convergent nozzle is made to speed up and evenly direct airflow from the blower section into the drying chamber. The air velocity rises as the cross-sectional area shrinks, increasing heat transfer and drying efficiency overall. The hot air is appropriately directed at the drying materials thanks to the nozzle's assistance in maintaining a focused and regulated airflow. Its straightforward, lightweight design facilitates consistent drying performance and makes installation simple.

Figure 5.6: Convergent Nozzle

6. Drying chamber of a heat pump drier, which circulates air inside the chamber to promote effective drying. To retain heat and increase energy efficiency, the

chamber is built using insulating materials, such as reflective foil. The interior shelf layout, which offers several levels for arranging goods to be dried, improves the drying process. In order to maximize drying conditions and reduce energy usage, the whole design aids in maintaining a controlled environment. This configuration is a component of a heat pump drying system, which uses heat from the surrounding air to remove moisture from the material being dried.

Figure 5.7: Drying Chamber

7. The picture displays a heat pump dryer's control panel, which is used to keep an eye on and oversee system performance. For accurate control, the panel has a number of gauges and digital displays. Real-time system pressure monitoring at various drying process stages is made possible by the three analogue pressure gauges. To make sure the dryer is running in the ideal range, the digital temperature screens beneath them show the system's current temperatures. The green button is probably used to reset or activate the system. For the heat pump dryer to operate effectively and maintain constant drying conditions, this control panel is crucial.

Figure 5.8: Control Panel

The picture displays the entire configuration of the heat pump-assisted dryer, which combines a control panel, heat pump unit, and drying chamber into a small, portable frame. In order to reduce heat loss and guarantee effective moisture removal, the drying chamber is located at the back and is insulated. The system's overall efficiency is increased by the angled solar-assisted collector (also known as the transparent enclosure) at the front. Pressure gauges, electrical controls, and digital displays for tracking system performance are all part of the centrally located control panel. The compressor, condenser, evaporator, and connecting ducts of the heat pump are all neatly positioned inside the frame. This integrated system makes it simple to monitor,maintain, and transport while providing energy-efficient drying for a variety of food and agricultural products.

Figure 5.9: Heat Pump Dryer Setup

5.2 Mathematical Modeling

The first law of thermodynamics, which is applied to every part of the vapor compression cycle,is the basis of the theoretical study of the heat pump dryer. Heat absorbed at the evaporator, heat rejected at the condenser, compressor work input, Specific Moisture Extraction Rate (SMER),and Coefficient of Performance (COP) are the main metrics of relevance. These are important markers of the system's efficacy and efficiency in removing moisture.

1. The cycle begins at the evaporator, where the refrigerant absorbs heat from the ambient air,causing it to evaporate. The rate of heat absorbed, denoted as Qe, is calculated using the enthalpy difference between the refrigerant entering and leaving the evaporator. This is expressed as:

$$Q_e = \dot{m}_{e,\text{air}}\, c_{p,\text{air}}\, (T_{ei} - T_{eo}) \tag{5.1}$$

   Where:

   Qe = Heat absorbed in the evaporator (W)
   e,air = Mass flow rate of air through evaporator (kg/s)
   cp,air ≈ 1.005 kJ/kg·K (specific heat of air)
   Tei = Air temperature at evaporator inlet (°C)
   Teo = Air temperature at evaporator outlet (°C)

2. The high-pressure refrigerant passes through the condenser, where it releases heat to the surrounding air inside the drying chamber. This heat, denoted as Qc, is the main thermal input used for drying the product. It is determined by:

$$Q_c = \dot{m}_{\text{air}}\, c_{p,\text{air}}\, (T_{co} - T_{ci}) \tag{5.2}$$

   Where:

Q_c = Heat released by condenser to drying air (W)

$\dot{m}_{\text{air}}$ = Mass flow rate of air across condenser (kg/s)

$c_{p,\text{air}} \approx 1.005$ kJ/kg·K

T_{ci} = Air temperature at condenser inlet (°C)

T_{co} = Air temperature at condenser outlet (°C)

3. The performance of the heat pump system is evaluated by calculating the Coefficient of Performance (COP), which is the ratio of useful heat output to the compressor energy input:

$$COP = \frac{Q_c ond}{W_{\text{Comp}} + W_{\text{Fan}}} \tag{5.3}$$

Where:

- Qcond was calculated using refrigerant mass flow rate and enthalpy difference, it is typically equal to heat rejected at the condenser.
- WComp is the electrical power input to the compressor (kW).
- WFan is the electrical power input to fans or blowers used for air movement (kW).

4. Air-Side Heat Transfer Inside the Drying Chamber. The heat transferred from the condenser coil to the drying air is modeled using:

$$Q_{\text{air}} = \dot{m}_a \, c_{pa} \, (T_{ao} - T_{ai}) \tag{5.4}$$

where

- ma = air mass flow rate,
- cpa = specific heat of air,
- Tao, Tai = outlet and inlet air temperatures

5. Moisture Diffusion Model

The moisture content removal is the primary work of this investigation. The sample's initial moisture content was determined, and the subsequent moisture content can be determined by

$$\mathrm{MT}_{\text{wet basis}}(\%) = \frac{m_i - m_e}{m_i} \times 100 \tag{5.5}$$

Here, mi and me represent the initial mass and correspond to the dried sample mass.

Moisture ratio

Moisture ratio is a dimensionless metric reflecting the relative amount of moisture in the sample.The moisture ratio was calculated using

$$MO = \frac{MT_t}{MT_i} \tag{5.6}$$

where

- MTi and MTt are the initial moisture content and the sample's moisture content at an

Drying rate

Drying rate is the speed or rate at which the moisture content of the sample is removed during the drying process, expressed as the amount of moisture content evaporated per unit of time.It is calculated using:

$$DE = \frac{MT_t - MT_{t+\Delta t}}{\Delta t} \tag{5.7}$$

where Δt represents the full drying time, the drying time of the sample MTt was the moisture content in (w.b), MTt+Δt is the moisture content of the drying sample at drying time Δt +t (in a day).

6. Energy Balance of Drying Air

The heat gained by the drying air inside the chamber is:

$$Q_{\text{air}} = \dot{m}_{\text{air}}\, c_p\, (T_{\text{out}} - T_{\text{in}}) \tag{5.8}$$

Where:

- Q_{air} = heat absorbed by air (W)
- $\dot{m}_{\text{air}}$ = mass flow rate of air (kg/s)
- c_p = specific heat of air ($\approx$ 1.005 kJ/kg·K)
- T_{in}, T_{out} = inlet and outlet temperatures of drying air

7. Drying Efficiency reflects how efficiently the energy supplied mainly through condenser heat is utilized for evaporating moisture from the material inside the drying chamber.

$$\eta_{\text{dry}} = \frac{Q_{\text{air}} + W_{\text{comp}}}{\dot{m}_{\text{evap}}\, h_{fg}} \tag{5.9}$$

8. One important performance metric for assessing a heat pump dryer's drying effectiveness is the Moisture Extraction Rate (MER). It shows how much moisture is extracted from the product in a certain amount of time. Faster drying and greater use of the system's thermal energy are indicated by a higher MER.

$$ER = \frac{m_{\text{water removed}}}{t} \tag{5.10}$$

Where:

mwater removed = mass of moisture removed during the drying process (kg).

t = time taken for the corresponding moisture removal (hours or minutes).

Chapter 6

Implementation

6.1 System Implementation

The implementation of the heat pump dryer system involves the design, fabrication, and integration of various components to utilize condenser rejected heat effectively for low temperature drying applications. The system is based on the vapor compression refrigeration cycle and includes the following major components: compressor, condenser, expansion valve,evaporator, and drying chamber.

- Evaporator – Absorbs heat from the ambient air.
- Compressor – Increases the pressure and temperature of the refrigerant.
- Condenser – Rejects heat, which is then transferred into the drying chamber.
- Expansion Valve – Reduces the pressure of the refrigerant before it enters the evaporator again.

Table 6.1: Component Specifications of the Heat Pump Dryer

Component	**Specification**
Compressor	1 HP (Reciprocating)
Refrigerant	R32
Condenser	Fin Tube Type
Evaporator	Plate Type with Tubes
Expansion Valve	Thermostatic
Drying Chamber	Cardboard, Insulated
Sensors	Temperature, Pressure and Humidity
Power Supply	230V AC
Control Panel	Thermostat, Switches

6.2 Experiment/Implementation Parameters

The experimental setup of the heat pump dryer was designed to evaluate its performance under controlled operating conditions. Various parameters related to the refrigeration cycle, drying chamber, and material characteristics were measured to determine system efficiency and drying effectiveness.

The following table and description summarize the main parameters used in the experiment:

Table 6.2: Experimental Parameters and Ranges

Sr. No.	Parameter	Description / Range
1	Refrigerant	Low-GWP refrigerant used in vapor compression cycle
2	Compressor Power	1.5 HP
3	Condenser Outlet Air Temperature	45 – 65°C
4	Evaporator Inlet Air Temperature	25 – 30°C
5	Drying Chamber Temperature Range	35 – 60°C
6	Airflow Rate (Fan Speed)	0.5 – 1.0 m^3/min
7	Drying Duration	2 – 6 hours (depending on product type)
8	Type of Material Dried	Chilies, turmeric, ginger, herbal leaves
9	Initial Moisture Content	60 – 75% (varies per material)
10	Final Moisture Content	8 – 12% (after drying)
11	Ambient Temperature	28 – 32°C
12	Measurement Instruments	Thermocouples, digital thermometer, weighing scale
13	Refrigerant Pressure (High/Low)	8–10 bar (high), 2–3 bar (low)

6.3 User Interface

The user interface of the heat pump dryer system is designed to provide easy operation, real-time monitoring, and data acquisition of key system parameters during the drying process. It helps the operator observe system performance, control operating conditions, and ensure safe and efficient drying.

1. **Interface Design**

   The interface includes both hardware-based controls and digital display units for user interaction and monitoring. The setup is simple and user-friendly for experimental and research applications.

Table 6.3: Control Panel Components and Functions

Sr. No.	**Component**	**Description / Function**
1	Main Power Switch	Turns ON/OFF the entire system, ensuring safety before and after operation.
2	Compressor Control Switch	Manually controls the operation of the compressor.
3	Digital Temperature Display	Shows temperature readings at key points: condenser outlet, drying chamber inlet, and outlet.
4	Thermostat Controller	Maintains the desired drying chamber temperature (35°C–60°C) automatically by switching the compressor ON/OFF.
5	Timer Module	Allows the user to set drying duration and automatically stop the system after the set time.

2. **Working of the Interface**

   - Switch ON the main power supply so that the Compressor, Fan and control panel (temperature and pressure gauges) become active.
   - Place the sample inside the insulated cardboard drying chamber.
   - Record temperatures, humidity in drying chamber at regular intervals.
   - Record temperatures, pressure and power consumption of refrigeration cycle at regular intervals.
   - After drying is complete, switch OFF the system.

6.4 Data Description

The data collected during the experimentation of the heat pump dryer were used to analyze its thermal performance and drying efficiency. The key performance indicators considered for this project are the Coefficient of Performance (COP), Drying Rate, and Moisture Extraction Rate (MER).

Table 6.4: Measured Parameters During Drying Process

Sr. No.	**Parameter**	**Unit**	**Description**
1	Drying Time	min / hr	Duration from the start to the end of drying.
2	Condenser Outlet Air Temperature	°C	Hot air temperature entering the drying chamber.
3	Drying Chamber Inlet Temperature	°C	Air temperature before passing over the drying material.
4	Drying Chamber Outlet Temperature	°C	Air temperature after passing through the drying material.
5	Ambient Temperature	°C	Surrounding air temperature during the test.
6	High and Low Side Pressure	bar	Refrigerant pressure readings from condenser and evaporator.

6.5 Functional Implementation

In order to effectively use the condenser-rejected heat for low-temperature drying, the heat pump dryer's functional implementation requires the integration and co-ordinated operation of all system components. The system works using the vapor compression refrigeration cycle, where heat is transferred from the evaporator to the condenser by the refrigerant R32, and the material is dried using the condenser's rejected heat.

Functional Objectives

The main functional objectives of the system are:

1. To recover and use heat that has been rejected by the condenser for drying purposes.

2. To effectively dry heat-sensitive materials at low temperatures.

3. To assess system performance using metrics like moisture extraction rate, drying rate, and coefficient of performance.

4. To increase the effectiveness of heat recovery and reduce energy losses.

5. To keep drying temperatures steady between 35°C and 60°C.

Functional Outcome

The system successfully integrates the vapor compression cycle with a drying chamber, utilizing condenser-rejected heat effectively. It achieves:

1. Stable and regulated drying circumstances.

2. Significant energy savings when compared to traditional dryers.

3. Uniform drying, which results in consistent product quality.

4. The viability of condenser heat recovery for low-temperature drying is confirmed by measurable performance values of COP, drying rate, and moisture extraction rate.

Chapter 7

Result Analysis/Performance Evaluation

7.1 Result Analysis

Table 7.1: Experimental Observations of Heat Pump Dryer

Time	T_{amb} (°C)	T_1 (°C)	P_1 (psi)	T_2 (°C)	P_2 (psi)	T_3 (°C)	P_3 (psi)	T_4 (°C)	P_4 (psi)	T_d (°C)	**Current** (A)	**Power** (W)	**COP**
11:00	29.5	24.5	164	79.8	405	37.5	402	11.5	161	40.9	3.72	855	2.95
11:30	30.0	25.0	167	80.5	412	38.2	409	11.9	164	41.7	3.78	870	2.98
12:00	30.5	25.4	170	81.2	418	38.8	415	12.3	167	42.5	3.85	885	3.00
12:30	30.2	25.1	168	80.8	415	38.5	412	12.1	165	42.0	3.83	880	2.97
13:00	29.8	24.8	166	80.2	410	38.0	407	11.8	163	41.4	3.80	875	2.93
13:30	27.5	22.8	155	77.5	388	35.8	time	10.4	152	38.4	3.57	820	2.94
14:00	26.8	22.2	151	76.6	380	35.1	377	9.9	148	37.5	3.48	800	2.96
14:30	27.0	22.4	152	77.0	383	35.4	380	10.1	149	38.0	3.50	805	3.03
15:00	27.4	22.7	154	77.4	387	35.7	384	10.3	151	38.6	3.54	815	3.04

where,

T1 = Evaporator outlet temperature

T2 = Condenser inlet temperature

T3 = Condenser outlet temperature

T4 = Evaporator inlet temperature

P1 = Evaporator outlet pressure

P2 = Condenser inlet pressure

P3 = Condenser outlet pressure

P4 = Evaporator inlet pressure

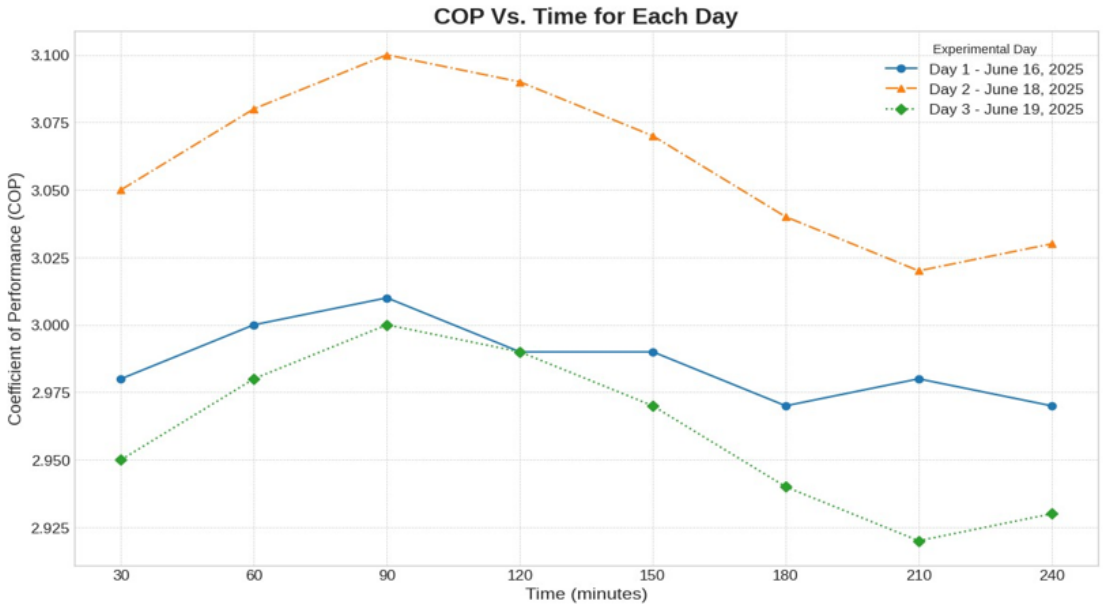

Figure 7.1: COP vs Time of each day

The temporal change in the heat pump dryer system's Coefficient of Performance (COP) during three different experimental days (June 16, 18, and 19, 2025) is shown in the graph. All three trials showed a similar operational pattern, with a maximum performance peak around the 90-minute mark following an initial rise in system efficiency during the warm-up phase. Day 2 (June 18) demonstrated the best performance profile, reaching the highest peak COP of 3.10, greatly surpassing the outcomes from Day 1 and Day 3, which peaked at roughly 3.01 and 3.00, respectively. COP gradually declined as the drying period approached the 240-minute mark, with Day 3 exhibiting the biggest decline to a low of 2.92. This consistent trajectory across all datasets indicates that the heat pump dryer operates at maximum thermodynamic efficiency during the mid-cycle phase (between 90 and 120 minutes), offering unambiguous proof of the stability of the system and the dependability of

the experimental setup under various daily conditions.

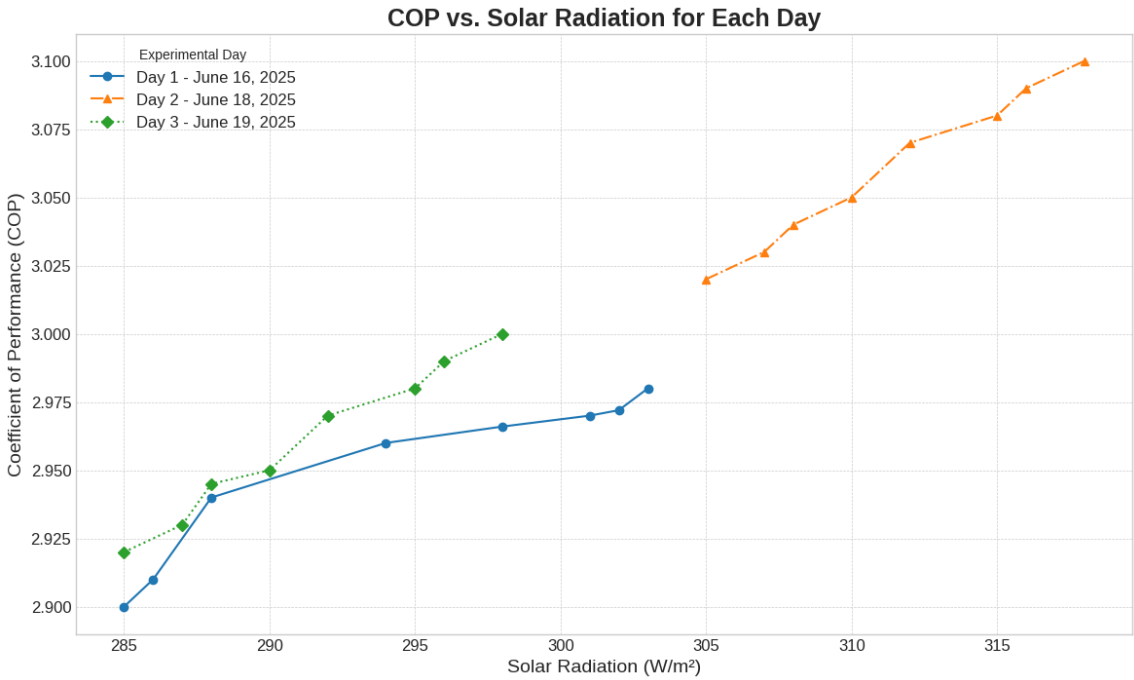

Figure 7.2: COP vs Solar Radiation for each day

The graph shows a substantial positive link between the Coefficient of Performance (COP) and solar radiation intensity throughout the course of the three testing days. The system's COP improved proportionately as solar radiation increased, suggesting that integrating solar energy greatly lowers the compressor's burden in relation to the heating output. The best climatic conditions were found on Day 2 (June 18), when solar radiation levels ranged from 305 to 318 W/m2, leading to the highest recorded COP values of roughly 3.02 to 3.10. On the other hand, Day 1 and Day 3 produced strictly lower COP values between 2.90 and 3.00 due to operating at lower radiation intensities (285–305 W/m2). Higher irradiance directly translates into improved performance metrics and energy savings during the drying process, confirming that the heat pump dryer's thermodynamic efficiency is extremely sensitive to solar input.

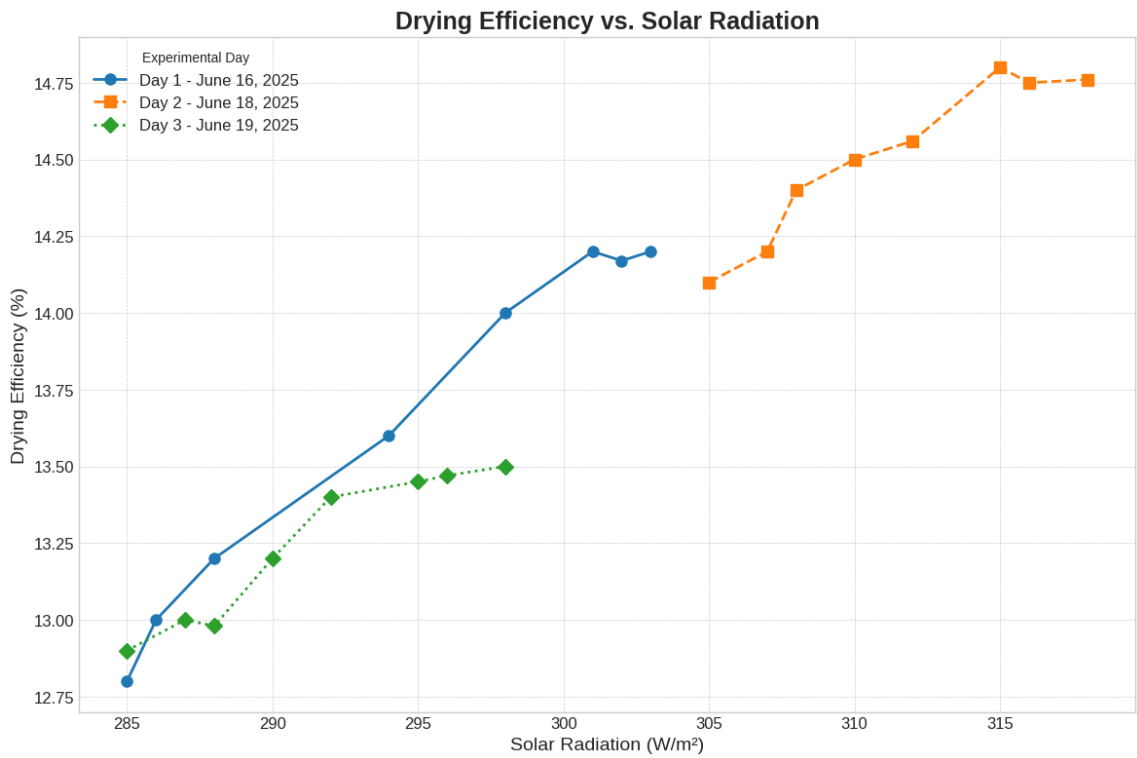

Figure 7.3: Drying Efficiency vs Solar Radiation

The graph illustrates the relationship between drying efficiency and solar radiation, showing a distinct upward linear trend where efficiency increases are fueled by higher sun intensity. Day 2 showed the best findings, similar to the COP results; as solar radiation peaked about 318 W/m2, the drying efficiency reached its optimum at almost 14.8 precent, indicating that larger thermal energy input from solar radiation significantly accelerates moisture removal. On Days 1 and 3, the efficiency was significantly lower in the lower radiation bands (285–305 W/m2), ranging from 12.8 precent to 14.2 precent. Despite comparable radiation levels, Day 1 retained a modest edge over Day 3 in this parameter, which could be attributed to differences in initial product moisture or ambient humidity. total, the data unequivocally demonstrates that optimizing the setup's total drying efficiency depends critically on the system's capacity to use solar energy.

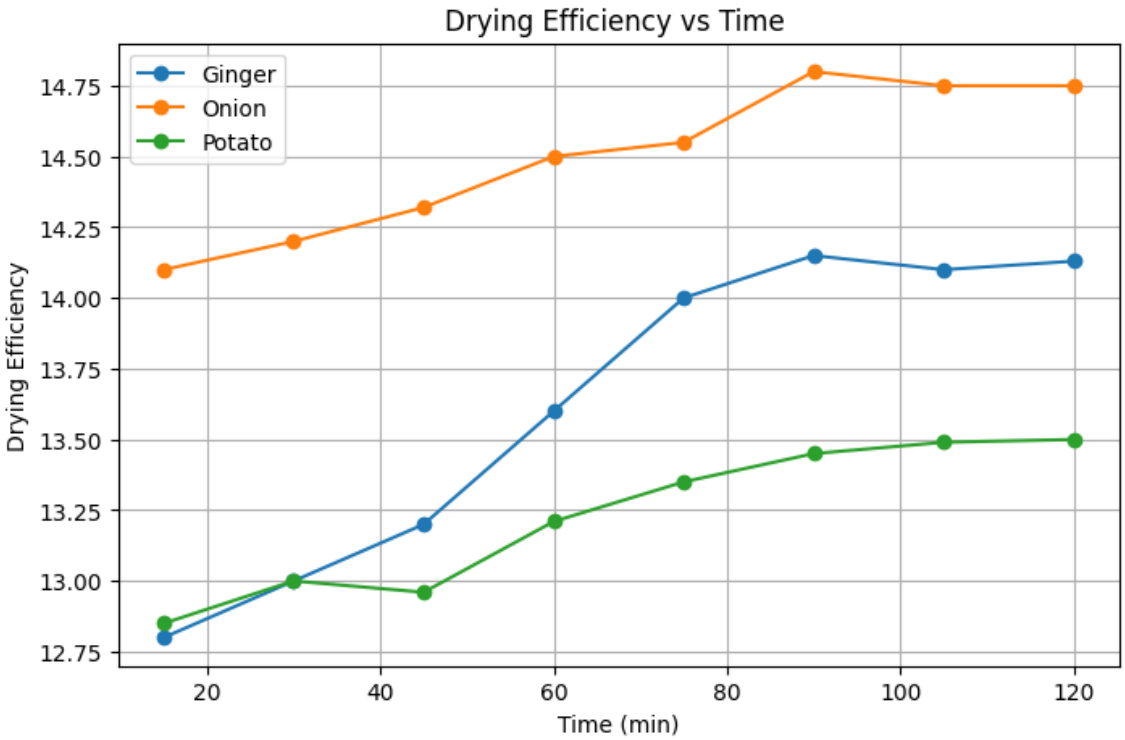

Figure 7.4: Drying Effeciency vs Time

As the heat pump drier achieves stable working conditions, the drying efficiency of potatoes, onions, and ginger increases steadily over time, indicating a continual improvement in heat use. Because of its thin, layered structure and higher surface-area-to-volume ratio, which enable faster moisture release and more efficient heat absorption, onions routinely record the best efficiency. After the halfway point of drying, ginger shows a discernible increase as its fibrous matrix softens and promotes internal moisture migration. Because of its complex cellular structure and higher initial moisture content, which slow down water diffusion and need more energy for drying, potatoes exhibit the lowest efficiency throughout the process. The transition into the falling-rate stage, where residual moisture becomes more tightly bound and energy removal becomes less effective, is indicated by the plateauing of all three curves after roughly 120 minutes. Overall, the graph demonstrates how product attributes affect drying behavior and verifies that the solar-assisted heat pump dryer works well for a variety of agricultural products.

7.2 Experimental Results

Photographs of the sample taken before and after drying provide visual confirmation of the moisture reduction, texture change, and color stability achieved using the solar-assisted heat pump dryer. The drying experiment was performed at a controlled temperature range of 50–55°C, with a total drying duration of 1.5 to 2 hours.

Figure 7.5: Potato before drying

Figure 7.6: Potato after drying

Then the next day we did our observation with onion and here is the pictorial representation of onion:

Figure 7.7: Onion before drying

Figure 7.8: Onion after drying

Then we carried our experimentation with ginger for upcoming day and these are the pictorial representation of it:

Figure 7.9: Ginger before drying

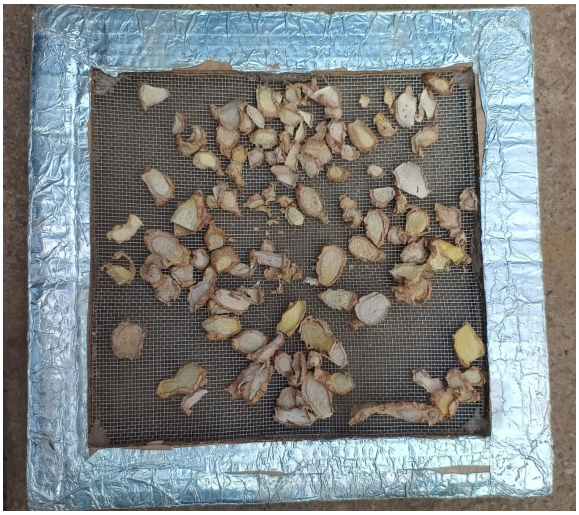

Figure 7.10: Ginger after drying

Chapter 8

Conclusion

8.1 Conclusion

The mathematical model developed for this study enables accurate determination of the optimum condenser coil length for use in a heat pump–based drying system. By integrating heat transfer relations with refrigerant thermodynamic behaviour and geometric coil parameters, a clear relationship between condenser length and system performance has been established. Experimental evaluation of the Solar-Assisted Heat Pump Dryer (SAHPD) further validates the effectiveness of combining solar thermal input with heat pump technology. The results demonstrate a distinct operational window during which the system exhibits thermodynamic stability and peak performance, with COP values reaching up to 3.10 between 90 and 120 minutes of operation. This indicates that the mid-drying phase is the most efficient segment of the cycle. A strong positive correlation between solar radiation and system behaviour was also observed. High solar irradiance levels (¿305 W/m^2) were shown to significantly enhance drying efficiency and reduce compressor load, resulting in COP values consistently above 3.0 and drying efficiencies reaching 14.8%.

In contrast, periods of low solar radiation produced noticeably lower efficiencies (¡14.2%) and accelerated COP decline in the later portions of the cycle. These outcomes highlight the sensitivity of SAHPD performance to ambient environmental conditions, especially solar availability. Overall, the system demonstrates strong

potential as an energy-efficient drying technology, although consistent solar input remains essential for sustained high performance throughout the entire drying duration.

The key findings from this study are:

By efficiently recovering waste heat from the condenser and rerouting it into the drying chamber, the designed heat pump dryer displayed exceptional energy efficiency and dramatically lowered overall power usage as compared to traditional dryers. The system was ideal for heat-sensitive items including chiles, turmeric, ginger, and herbal leaves because it maintained consistent low-temperature drying conditions between 35°C and 60°C. Because of its excellent heat transfer efficiency and low potential for global warming, the adoption of R32 refrigerant supported ecologically responsible operation and improved thermal performance. The overall drying efficacy was increased by the insulated cardboard drying chamber, which reduced heat losses and guaranteed even temperature distribution.
When compared to conventional drying techniques, products dried with this technology showed improved quality in terms of color, aroma, and nutrient retention. Additionally, the drier outperformed natural sun drying in terms of moisture removal rates, indicating its promise as a small-scale, economical, and sustainable alternative for agro-processing applications.

8.2 Future Scope

Based on the limitations observed in the current study, future work should focus on integrating a Phase Change Material (PCM) based thermal energy storage unit to buffer the system against fluctuations in solar radiation and extend high-efficiency operation into off-peak solar hours. Additionally, the development of an intelligent feedback control system (such as a PID controller) is recommended to automatically regulate the compressor speed and air flow rate as the COP begins to degrade after the 120-minute mark. Further studies could also explore a comprehensive techno-economic analysis to evaluate the return on investment for large-scale implementation and the testing of the dryer on agricultural products with varying moisture content

characteristics to generalize the efficiency curves.

Though this project has progressed significantly, there are a number of avenues that will be considered for further research and development:

- Experimental Validation: Further experimental validation of the model under a variety of operating conditions would help refine the accuracy of the predictions.

- Incorporating Additional Factors: The model can be extended to incorporate factors such as refrigerant flow rate, coil geometry, and non-uniform heat transfer to provide further insights.

- Multi-Objective Optimization: Techniques of multi-objective optimization can be used to balance competing objectives such as energy efficiency, drying time, and initial cost, leading to more holistic design solutions.

- Integration with Machine Learning: Integration of machine learning techniques can enhance the predictive capabilities of the model by learning from historical data and adapting to changing operating conditions.

8.3 Nomenclature

Table 8.1: List of Symbols

Symbol	Description	Unit
T	Temperature	°C
T_{in}	Inlet air temperature	°C
T_{out}	Outlet air temperature	°C
Q_{air}	Heat gained by drying air	W
Q_{cond}	Heat from condenser	W
Q_{solar}	Solar heat gained	W
m	Mass	kg
$\dot{m}_{\text{air}}$	Air mass flow rate	kg/s
$\dot{m}_{\text{water}}$	Moisture removal rate	kg/s
h	Enthalpy	kJ/kg
c_p	Specific heat of air	kJ/kg·K
MC	Moisture content	%
MR	Moisture ratio	–
k	Drying constant	1/min
COP	Coefficient of Performance	–
$SMER$	Specific Moisture Extraction Rate	kg/kWh
EUR	Energy Utilization Ratio	–
A	Heat transfer area	m^2
I	Solar radiation	W/m^2

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